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LIBRARY IEEE;
USE IEEE.STD_LOGIC_1164 .ALL;

PACKAGE my_conversion_package IS
    FUNCTION ToChar(StdLogicValue : STD_LOGIC) RETURN CHARACTER;
    FUNCTION ToSTRING( VectorValue : STD_LOGIC_VECTOR ) RETURN STRING;
    FUNCTION ToChar(int: INTEGER) RETURN CHARACTER;
    FUNCTION ToSTRING(int: INTEGER) RETURN STRING;
END my_conversion_package ;


PACKAGE BODY my_conversion_package IS

   FUNCTION ToChar( StdLogicValue : STD_LOGIC)
         RETURN CHARACTER IS
   BEGIN
         CASE StdLogicValue IS
               WHEN 'U' => RETURN 'U';
               WHEN 'X' => RETURN 'X';
               WHEN '0' => RETURN '0';
               WHEN '1' => RETURN '1';
               WHEN 'Z' => RETURN 'Z';
               WHEN 'W' => RETURN 'W';
               WHEN 'L' => RETURN 'L';
               WHEN 'H' => RETURN 'H';
               WHEN '-' => RETURN '-';
               WHEN OTHERS => RETURN 'X';
         END CASE;
   END ToChar;

   FUNCTION ToChar(int: INTEGER) RETURN CHARACTER is
         VARIABLE c: CHARACTER;
   BEGIN
         CASE int IS
               WHEN 0 => c := '0';
               WHEN 1 => c := '1';
               WHEN 2 => c := '2';
               WHEN 3 => c := '3';
               WHEN 4 => c := '4';
               WHEN 5 => c := '5';
               WHEN 6 => c := '6';
               WHEN 7 => c := '7';
               WHEN 8 => c := '8';
               WHEN 9 => c := '9';
               WHEN OTHERS => c := '?';
         END CASE;
         RETURN c;
   END ToChar;

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   FUNCTION ToSTRING(int: INTEGER) RETURN STRING IS
         VARIABLE str: STRING(9 DOWNTO 1); -- STRING's lowest index must be 1
         VARIABLE n: INTEGER;
         VARIABLE index : INTEGER := 1;

   BEGIN
         n := ABS(int);

         IF ( n = 0 ) THEN
               str(1) := '0';
               index := 1;
         ELSE
            WHILE (n /= 0) LOOP
               str(index) := ToChar( n MOD 10);
               n := n / 10;
               index := index + 1;
            END LOOP;
         END IF;
         IF int < 0 THEN
               RETURN '-'& str(index-1 DOWNTO 1);
         ELSE
               RETURN str(index-1 DOWNTO 1);
         END IF;
   END ToSTRING;


   FUNCTION ToSTRING(Vectorvalue : STD_LOGIC_vector )
             RETURN STRING IS
         VARIABLE OutSTRING : STRING(1 to Vectorvalue 'length);
         VARIABLE k : INTEGER;
   BEGIN
         k := 1;
         FOR i IN Vectorvalue 'length - 1 DOWNTO 0 LOOP
               OutSTRING(k) := ToChar(Vectorvalue (i));
               k := k+1;
         END LOOP;
         RETURN OutSTRING;
   END ToSTRING;

END my_conversion_package ;
```